

RESEARCH ARTICLE

Poor and rich dolphin optimization algorithm with modified deep fuzzy clustering for COVID-19 patient analysis

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Email: sudhagard70@gmail.com**Summary**

The Coronavirus disease 2019 (COVID-19) is considered as a pandemic by the World Health Organization (WHO), which has spread worldwide. Over millions of peoples are infected across the globe and several people are died. However, the most worrying group of patients suffered from lung severity with respiratory failure. Hence, cluster analysis is utilized for examining the heterogeneity of diseases for determining various clinical phenotypes having the same traits. This article devises an optimization-driven technique for COVID-19 patient analysis using the spark framework. Here, the input data is partitioned and fed to different slave nodes. In slave node, the selection of imperative features is done using the proposed poor and rich dolphin optimization algorithm (PRDOA). The proposed PRDOA is obtained by combining poor and rich (PRO) and dolphin echolocation (DE) algorithm. The fitness is newly devised considering Minkowski distance measure. The clustering is performed on the master node using the proposed Tanimoto-based deep fuzzy clustering (TDFC) for effective COVID-19 patient analysis. Thus, the proposed TDFC is obtained by incorporating Tanimoto concept and deep fuzzy clustering. The proposed PRDOA with TDFC offered enhanced performance with the highest clustering accuracy of 89.8%, dice coefficient of 90%, Jaccard coefficient of 85.7%, and rand coefficient of 85.7%.

KEYWORDS

Coronavirus disease 2019 patient analysis, deep fuzzy clustering, Minkowski distance, spark architecture, Tanimoto distance

1 | INTRODUCTION

The techniques of big data have acquired an imperative part in several factors that involve the quick collection of multiple-source big data, quick visualization of endemic data, spatial tracking of verified cases, forecasting of local communication, spatial segmentation of the pandemic risk and level of prevention, balancing and controlling the supplies and desires of resources, emotional guidance and elimination of panic and effectual analysis of COVID-19 anticipation, and control. Geographic information system (GIS) emerged quickly and poses whole technological paths for preparing the data, constructing the platform and model and producing the map. Here, the struggle of extensive pandemic causes several issues in discovering the classical techniques and enhance accuracy and speed of data provision in social management.¹⁻⁴ In big data, data no longer come from the government, but are acquired from most assorted enterprises.⁵ With big data analysis models, the data regarding the personal relations, locations of health centers, and the notorious techniques for the spread of Coronavirus is detected. These techniques also utilized computational models for predicting the disease spread using forward simulation considering a patient and infection propagation and backward simulation for tracking the

count of infections.⁶⁻⁸ For analyzing the influence of policies and healthcare economics, precise and feasible data are imperative. The official data regarding COVID-19 are not complete and contradictory. The reported values are far from mathematical modeling.⁹

COVID-19 was determined in 2019 end from Wuhan located in China.¹⁰ The virus has spread quickly by transmission all over the world in a short time-span. The World Health Organization (WHO) has to announce COVID-19 as pandemic. For eliminating the epidemic circumstances, various countries have adopted certain proposals for re-molding the quality of health care.⁷ Clustering of countries having the same quality of health care assists to learn regarding the health care quality in contrast to other countries. The clustering is considered as an imperative part and is extensively utilized in unsupervised machine learning techniques as it is extensively utilized for determining the hidden structure in an automated manner. In the domain of machine learning, both unsupervised and supervised learning are admired for solving different types of real-world issues.^{11,12} For clustering, unsupervised machine learning techniques are extensively utilized.¹³ The clustering technique discovers and splits data points based on similarity of hidden structures of the database. This technique is utilized for discovering the count of clusters as it is liable to split the unlabeled database into subsets. It is performed by evaluating distance among cluster centroids. In the classical K-means clustering technique, the initial centroids are chosen in a random manner. Hence, discovering the initial coordinate points of centroids is considered as an imperative part in clustering.¹⁴

The analysis of clusters^{15,16} is general statistical technique having victorious applications to COVID-19, and is most widely considered as epidemiology. With respect to group data points, the analysis of cluster is utilized for studying infectious diseases, non-communicable diseases and pandemic outbreaks, like SARS, Ebola, and COVID-19.¹⁷ Clustering techniques are altered and common instances are spectral clustering and K-means clustering that divides the attribute into discrete sets and hierarchical clustering that does not devise an accurate count of clusters.¹⁸⁻²⁰ Various techniques²¹ are devised for determining the initial centroids of cluster. The researchers devised a centroids selection technique on the basis of radial and angular coordinates. Here, the count of iterations varies for each instance. In addition, the time taken for implementing a technique elevates radically with the increasing number of clusters. Moreover, the small databases are utilized for simulation outcomes, which cannot offer insight on huge dataset. No time of execution was provided for the devised model.¹⁴

The goal is to present an optimization assisted technique based on clustering for analyzing the COVID-19 patient using the spark framework. Here, the patient's data is considered as an input which is further splitted into small subsets. Each subset of data is fed to set of slave nodes for selecting imperative features. In each slave node, the selection of the most significant feature is done using the proposed poor and rich dolphin optimization algorithm (PRDOA). The proposed PRDOA is generated by integrating poor and rich (PRO) and dolphin echolocation (DE) algorithm. The fitness is newly devised considering Minkowski distance measure. The clustering is carried out on the master node with the proposed Tanimoto-based deep fuzzy clustering (TDFC) for effectual COVID-19 patient analysis. Hence, the developed TDFC is generated by combining Tanimoto concept and deep fuzzy clustering (DFC).

The major contributions of this article are

- **Proposed PRDOA for effective feature selection:** The selection of imperative features is done in slave nodes using the partitioned data. Here, the features are selected using the proposed PRDOA, which is obtained by combining DE and PRO.
- **Proposed Tanimoto-based deep fuzzy clustering for patient data analysis:** The clustering is performed on master node with the proposed TDFC, which will be the integration of the Tanimoto concept²² and DFC.

The rest of the sections are described below: Section 2 describes the classical techniques for COVID-19 patient analysis. Section 3 presents the proposed model for COVID-19 patient data analysis. Section 4 describes the efficiency of the proposed model with respect to existing models. Section 5 is the conclusion.

2 | LITERATURE SURVEY

The eight classical techniques based on big data clustering for COVID-19 patient data analysis are described along with its issues. Alazab et al.²³ devised an artificial intelligence technique on the basis of deep convolutional neural network for discovering COVID-19 patients with real-world databases. The system attained chest x-ray images for identifying the patients. The findings represent that this kind of assessment was beneficial for COVID-19 diagnosis. However, the techniques failed to consider other parameters, like humidity, temperature, and terrain. In addition, the time consumption was very high. Marco-Franco et al.⁹ developed a scheme for COVID-19 that incorporates data based on mortality rate. The prediction of casualties in Spain, and the infectious-case fatality rate (IFR) and case fatality rate (CFR) for outbreak were evaluated. However, the technique failed to involve age-related fatality ratio. Kumar²⁴ developed a cluster assessment based on data mining for classifying a real set of infectious diseases. The outcomes obtained permitted to sense the clusters of affected regions. The aim was to optimize the monitoring methods in affected regions that are beneficial to doctors and government in understanding the seriousness of COVID-19. This technique sometimes generated error values. James and Menzies²⁵ devised an assessment that resulted to cluster countries based on active cases with Johns

Hopkins epidemiological data. The obtained result of the cluster was helpful for various policy makers like managers and physicians. In addition, the clustering technique was adapted for comparing several COVID time-series data. However, this technique failed to deal with real-world applications. Villanustre et al.⁶ developed a technique for analyzing the COVID-19 patient data. This method helped among several domains, like healthcare, medical, and public applications. The developed technique handled the solutions for solving public health issues. However, the consumption of time was high. Zoabi et al.²⁶ developed a machine learning technique, which trained records using 51,831 tested individuals. The test set comprises data among the consequent week. The model helped to predict COVID-19 test results with elevated accuracy considering binary features. The method can be utilized among other considerations for prioritizing testing of COVID-19, while testing resources were less. However, this technique failed to generate improved outcomes. Prinja et al.²⁷ designed susceptible-exposed-infectious-recovered (SEIR) for predicting the health results using an unmitigated scenario that consists of air travel restrictions, and the present case consists of air travel restrictions among 8-week lockdown. Moreover, the technique computed the efficiency of lockdown. The computational complexity of this model was high. Zubair et al.¹⁴ developed an effective K-means clustering technique that discovered the initial centroids in an effective manner. The technique helped to discover the quality of health care using COVID-19 databases. The technique showed that the model reduced the count of iterations and time of execution for analyzing COVID-19. However, this technique failed to process with large instances.

2.1 | Challenges

The issues faced by big data clustering for COVID-19 patient data analysis is described below.

- In Reference 23, it is observed that the most infected regions have similar features and the spread of disease is high in coastal areas as compared to non-coastal areas. Here, the challenge exists in examining the impacts of humidity, temperature, and terrain in COVID-19 cities.
- Deep learning-assisted prediction techniques are utilized for predicting the count of COVID-19 confirmations, and recoveries with accuracy. However, the issues rely on examining these aspects with several advanced prediction techniques for effectual accuracy.²³
- The indicator of COVID-19 can be utilized for predicting the imminent inclination of COVID-19 and has its own issues that cannot illustrate other aspects, which can affect the weather and mobility. Hence, the issue relies on devising a strong model for predicting the trend with inclusive features.⁶
- The issue in SEIR mathematical model is reproduction count that alters among the states of India that owns to the various density of population, individual movement, demographic composition that are likely to impact the disease severity. Thus, the issue relies on producing epidemiological evidence and performing model-based assessment at the state level for informing local policies.²⁷
- K-means clustering technique outperformed in real-world cases as it can be diverged if the count of instances is high. Hence, the issue relies in planning work with large instances.

3 | PROPOSED TANIMOTO-BASED DEEP FUZZY CLUSTERING FOR COVID-19 PATIENT ANALYSIS

COVID-19 is considered as a global disease and is termed as a pandemic by WHO, wherein the virus is spread from China to whole world. There is more than million cases that suffer from COVID-19 and greater fatalities, because of COVID-19. The factors that incline a person to a high risk of death from COVID-19 are still not understood. Thus, a COVID-19 patient analysis is essential to enhance patient care and deploy healthcare resources throughout this pandemic. The goal is to devise a technique, namely, Tanimoto-based deep fuzzy clustering with hybrid optimization for analyzing COVID-19 patients. The analysis of clusters is done to categorize to several states of COVID-19. COVID-19 results in a global pandemic with serious human, economic, and social costs. For managing economic ramifications or prioritize safety of citizens, the policy-makers have sought multilevel technique that includes social distancing, movement restrictions and business closures. Hence, a precise discovery regarding the COVID-19 spread is the most imperative task. The steps undergone in COVID-19 patient analysis include feature selection and clustering. The COVID-19 patient analysis is performed with the spark model, which consists of the master node and slave node. At first, the input data is subjected to feature selection, which is performed in the slave node to choose relevant features. Here, the selection of features is done using proposed PRDOA, which is developed by combining PROA²⁸ and DE²⁹ algorithms. After selecting the imperative features, the clustering is done in master node with the proposed Tanimoto-based deep fuzzy clustering, which is obtained by combining Tanimoto concept²² and DFC.³⁰ Figure 1 presents a schematic view of developed Tanimoto-based deep fuzzy clustering with proposed PRDOA for COVID-19 patient analysis.

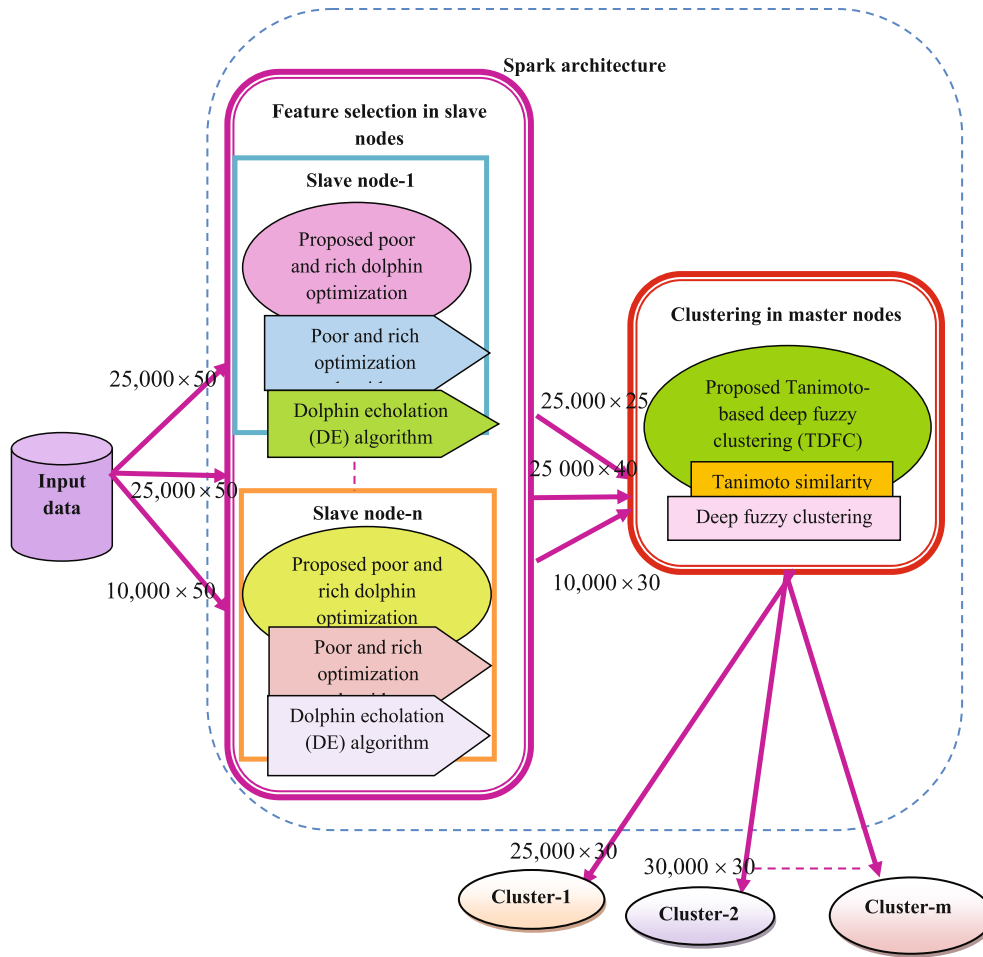


FIGURE 1 Architecture of developed Tanimoto-based deep fuzzy clustering for COVID-19 patient analysis

Imagine an input data be given as A that considers different attributes, and provided as

$$A = \{G_{r,s}\}; (1 \leq r \leq R); (1 \leq s \leq S), \quad (1)$$

where $G_{r,s}$ indicates data contained in dataset A depicting sth attribute of r th data, the total data points are represented by R , and total attributes are denoted by S .

The input data $G_{r,s}$ is given to spark architecture, wherein the spark model produces elevated processing power as the slave node server contains the potential to run parallelly. Initially, the partitioning of input data is done, and the partitioned data is fed to various slave nodes. Here, the features are chosen in the slave nodes.

The expression for partitioning the dataset into numerous subsets is

$$G_{r,s} = \{F_q\}; (1 \leq q \leq Q), \quad (2)$$

where Q signifies the totally generated partitions.

Slave nodes in spark:

The slave nodes and total data subsets are same, and in slave node the features are extracted from every partitioned data. Thus, the input to the m th slave node is expressed as

$$F_q = \{K_{m,n}\}; (1 \leq m \leq M); (1 \leq n \leq N), \quad (3)$$

where $K_{m,n}$ signifies q th partition data with m th attribute of n th data. Here, each splitted data is given to the number of slave nodes for selecting imperative features. Thus, the slave nodes perform feature selection with the proposed PRDOA algorithm, which is described in the subsection.

3.1 | Feature selection in slave nodes using proposed PRDOA algorithm

The feature selection offers a way to reduce the time of computation, enhance performance and offer an improved understanding of data. In addition, the feature selection is effective in making the data suitable for further processing. The goal of feature selection is building a simple and comprehensible model with clean and understandable data. Moreover, the feature selection process is considered as an imperative step to speed up learning and enhance the quality of data. Here, the feature selection is done using the proposed PRDOA algorithm, which is obtained by combining PRO²⁸ and DE²⁹ algorithms. The solution encoding, and fitness function steps of the proposed PRDOA algorithm are described below.

3.1.1 | Solution encoding

The solution representation is imperative to discover the best solution in order to solve optimization issues. For COVID-19 patient analysis, the solution set comprises set of selected features, which is obtained using the proposed PRDOA and fitness factor. Hence, based on fitness, the solution set acquires optimal features. Figure 2 portrays solution representation for best feature selection with developed PRDOA. Here, k expresses the index of features using the proposed PRDOA and fitness factor.

3.1.2 | Fitness function

The evaluation of fitness is done using Minkowski distance. Here, the distance is evaluated among selected features and class labels. Thus, the fitness function is formulated as

$$\text{Fitness} = \min (\zeta (k_j, n_j)), \quad (4)$$

where $\zeta(\cdot)$ signifies Minkowski distance, k_j refers selected feature, and n_j is class label. The Minkowski distance helps to find similarity among two real-valued functions and is useful to increase speed and accuracy. In addition, it is helpful for computing the distance among two data points in a multivariate space. The Minkowski distance refers a metric that evaluates distance among two real-valued vectors, which is termed as a generalization of both Manhattan and Euclidean distance. The Minkowski distance ζ is expressed as

$$\zeta_{kn} = \left[\sum_{d=1}^{\varpi} |D_k^d - D_n^d|^\alpha \right]^{1/\alpha}, \quad (5)$$

where α is integer, D_k^d refers to the selected features variable, and D_n^d refers class label variable. Here, $\alpha = 1$ refers the Mannhattans distance and $\alpha = 2$ signifies Euclidean distance. It is considered as a minimization function.

3.1.3 | Proposed PRDOA algorithm

The PRO²⁸ is motivated from the efforts of two sets of people who are poor, and rich who tries to acquire wealth. The technique produces acceptable solutions to real-world issues. This algorithm converges with few functions, and is capable to address several engineering problems. The PRO is adapted as a suitable choice for addressing optimization issues. Meanwhile, DE²⁹ is motivated from the mimicking behavior of dolphins that perform hunting. The DE works based on computational effort in which the user affords for optimization. Here, the DE provides a suitable convergence factor based on the problem. The DE provides the optimum answer in the count of loops offered by the user. The combination of PRO and DE helps to enhance overall performance with global optimal solution. The steps involved in the proposed PRDOA are given below.



FIGURE 2 Solution representation using PRDOA algorithm

Step 1. Initialization

In this phase, the solutions and other attribute, like iteration count g are initialized. The solutions are expressed as

$$B = \{B_1, B_2, \dots, B_l, \dots, B_m\}, \quad (6)$$

where m signifies total solutions, B_l refers l th solution.

Step 2. Evaluation of fitness

In Section 3.1.2 the determination of the fitness has been explained.

Step 3. Discovering update equation

The major population of PRO²⁸ consist of two subpopulations, namely, poor and rich. Hence for each iteration, the position of member must be varied on the basis of rich and poor mechanisms.

(a) The change with position of poor member

The change in poor position is given as

$$\vec{B}_{\text{poor},d}^{\text{new}} = \vec{B}_{\text{poor},d}^{\text{old}} + \left[p(\text{Pattern}) - \vec{B}_{\text{poor},d}^{\text{old}} \right], \quad (7)$$

where $\vec{B}_{\text{poor},d}^{\text{old}}$ refers location of member of poor population at previous iteration, p is random number, and Patter denotes a pattern of getting rich. The pattern is formulated as

$$\text{Pattern} = \frac{\vec{B}_{\text{rich,best}}^{\text{old}} + \vec{B}_{\text{rich,mean}}^{\text{old}} + \vec{B}_{\text{rich,worst}}^{\text{old}}}{3}, \quad (8)$$

where $\vec{B}_{\text{rich,best}}^{\text{old}}$ refers location of the best member of the rich population at previous iteration, $\vec{B}_{\text{rich,mean}}^{\text{old}}$ denotes the average position of the rich population, and $\vec{B}_{\text{rich,worst}}^{\text{old}}$ expresses the position of a worst member considering the rich population.

(b) Mutation

A normal distribution having zero mean and variance of one and the mutation probability of poor and rich are carried out in a separate manner. Thus, the mutation of rich and poor population is described.

If $\text{rand} < \text{Nmut}$

$$\vec{B}_{\text{rich},b}^{\text{new}} = \vec{B}_{\text{rich},b}^{\text{new}} + \text{rand}j, \quad (9)$$

End

Where rand signifies arbitrary value between 0 and 1, Nmut refers mutation probability, $\vec{B}_{\text{rich},d}^{\text{new}}$ is position of new value before performing mutation on rich position, $\vec{B}_{\text{poor},d}^{\text{new}}$ symbolizes the position of new value before performing mutation on poor position, and $\text{rand}j$ refers normalized distribution of arbitrary number.

If $\text{rand} < \text{Nmut}$

$$\vec{B}_{\text{poor},d}^{\text{new}} = \vec{B}_{\text{poor},d}^{\text{new}} + \text{rand}j \quad (10)$$

End

(c) The change using position of rich member

The update rule of PRO is utilized for improving the search space. According to PRO,²⁸ the update position of each rich population is given as

$$\vec{B}_{\text{rich},d}^{\text{new}} = \vec{B}_{\text{rich},d}^{\text{old}} + p \left[\vec{B}_{\text{rich},d}^{\text{old}} - \vec{B}_{\text{poor,best}}^{\text{old}} \right], \quad (11)$$

where $\vec{B}_{\text{poor,best}}^{\text{old}}$ refers location of the poor population best member at the previous iteration, $\vec{B}_{\text{rich},d}^{\text{old}}$ is current value of d th position of rich people at previous iteration, and p signifies class gap parameter.

This can be written as

$$\vec{B}_{\text{rich},d}^{g+1} = \vec{B}_{\text{rich},d}^g + p\vec{B}_{\text{rich},d}^g - p\vec{B}_{\text{poor,best}}^g \quad (12)$$

$$\vec{B}_{\text{rich},d}^{g+1} = \vec{B}_{\text{rich},d}^g (1 + p) - p\vec{B}_{\text{poor,best}}^g \quad (13)$$

The DE is effectual in solving the optimization problems and provides better performance. As per DE,²⁹ the update of DE is given by

$$B_d^{g+1} = B_d^g + E_d^{g+1}, \quad (14)$$

where E_d^{g+1} refers new search space dimension of the dolphin, and B_d^g signifies the current dolphin position.

$$B_d^{g+1} = B_d^g + E_d^g + \alpha_{1d} (M_d - B_d^g) + \alpha_{2d} (D - B_d^g), \quad (15)$$

where E_d^g is the current search space dimension of dolphin, α_{1d} and α_{2d} depict random numbers in [0, 1] adapted on d th location of dolphin, M_d personal best position, and D refers global best position discovered by dolphin.

$$B_d^{g+1} = B_d^g [1 - \alpha_{1d} - \alpha_{2d}] + E_d^g + \alpha_{1d} M_d + \alpha_{2d} D, \quad (16)$$

$$B_d^{g+1} - E_d^g - \alpha_{1d} M_d - \alpha_{2d} D = B_d^g [1 - \alpha_{1d} - \alpha_{2d}], \quad (17)$$

$$B_d^g = \frac{B_d^{g+1} - E_d^g - \alpha_{1d} M_d - \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}}. \quad (18)$$

Assuming $B_{rich,d}^g = B_d^g$, then substitute Equation (18) in Equation (13),

$$\overrightarrow{B_{rich,d}^{g+1}} = \frac{B_{rich,d}^{g+1} - E_d^g - \alpha_{1d} M_d - \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) - p \overrightarrow{B_{poor,best}^g}, \quad (19)$$

$$\overrightarrow{B_{rich,d}^{g+1}} = \frac{B_{rich,d}^{g+1}}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) - \frac{E_d^g + \alpha_{1d} M_d + \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) - p \overrightarrow{B_{poor,best}^g}, \quad (20)$$

$$\overrightarrow{B_{rich,d}^{g+1}} - \frac{B_{rich,d}^{g+1}}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) = - \frac{E_d^g + \alpha_{1d} M_d + \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) - p \overrightarrow{B_{poor,best}^g}. \quad (21)$$

$$\overrightarrow{B_{rich,d}^{g+1}} \left[1 - \frac{(1 + p)}{1 - \alpha_{1d} - \alpha_{2d}} \right] = - \frac{E_d^g + \alpha_{1d} M_d + \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) - p \overrightarrow{B_{poor,best}^g}, \quad (22)$$

$$\overrightarrow{B_{rich,d}^{g+1}} \left[\frac{1 - \alpha_{1d} - \alpha_{2d} - 1 - p}{1 - \alpha_{1d} - \alpha_{2d}} \right] = - \frac{E_d^g + \alpha_{1d} M_d + \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) - p \overrightarrow{B_{poor,best}^g}, \quad (23)$$

$$\overrightarrow{B_{rich,d}^{g+1}} \left[\frac{\alpha_{1d} + \alpha_{2d} + p}{1 - \alpha_{1d} - \alpha_{2d}} \right] = \frac{E_d^g + \alpha_{1d} M_d + \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) + p \overrightarrow{B_{poor,best}^g}. \quad (24)$$

The final update equation of the proposed PRDOA algorithm is given by

$$\overrightarrow{B_{rich,d}^{g+1}} = \frac{1 - \alpha_{1d} - \alpha_{2d}}{\alpha_{1d} + \alpha_{2d} + p} \left[\frac{E_d^g + \alpha_{1d} M_d + \alpha_{2d} D}{1 - \alpha_{1d} - \alpha_{2d}} (1 + p) + p \overrightarrow{B_{poor,best}^g} \right]. \quad (25)$$

Step 4. Re-evaluate fitness of solutions

After completing the solution update, re-evaluation of the fitness is achieved using Equation (4). The maximal fitness is selected as the optimal solution.

Step 5. Termination

The termination is done using the stopping criterion, which is checked to indicate the optimization end. Table 1 depicts the pseudo code of developed PRDOA.

The features selected using the proposed PRDOA and fitness factor are denoted as k .

3.2 | Clustering using proposed Tanimoto-based deep fuzzy clustering algorithm in master nodes

The master node attains selected features k from the splitted data using slave nodes and performs clustering for COVID-19 patient analysis. The analysis of COVID-19 patients is done using the TDFC algorithm, which Tanimoto coefficient and DFC.³⁰ Here, the DFC model determines a deep representation and performs fuzzy clustering using a novel depiction. For facilitating positive memberships, a deep normalized fuzzy compactness and separation (DNFCS) is utilized, which is described as follows:

TABLE 1 Pseudo code of developed PRDOA

PRDOA
Input: Population B
Output: Optimal Solution B^*
Begin
Population initialization
Sort the population into rich and poor
While ($g < g_{\max}$)
Forrich solution
Select the rich and best poor
Update current position using Equation (25)
Determine the rich fitness using Equation (4)
If there is other rich
Update current position using Equation (25)
Else if
Store the solution in B
End if
End for
For poor solution
Select the poor
Evaluate the pattern
Update the current position using Equation (7)
If there is other rich
Update current position using Equation (7)
Else if
Store the solution in B
End if
End for
Merge all solutions
Sort the solution into poor and rich
Return best rich B^*
End

Assume $H = \{h_1, \dots, h_b, \dots, h_C\}$, and $h_b \in R^i$ signifies hidden illustration obtained from auto-encoder. This model also poses fuzzy clustering layers to acquire h_b as input and offer output fuzzy membership $f_{b,c}$. The fuzzy membership $f_{b,c}$ of DNFCs is given in which β refers assigned hyper-parameter for balancing within-cluster distance and between-cluster distance. Here, the Tanimoto distance is applied on fuzzy membership $f_{b,c}$ of DNFCs, such that the equation becomes

$$\ell(f_{b,c}, \mu_c) = \sum_{b=1}^C \sum_{c=1}^i f_{bc}^t s(h_b, \mu_c) - \sum_{b=1}^C \sum_{c=1}^i \eta_c f_{bc}^t s(h_b, \bar{\mu}), \quad (26)$$

where $f_{b,c}$ signifies fuzzy membership, h_b refers to the hidden feature, μ_c refers to the trained weights, $s(\cdot)$ is Tanimoto distance, η_c signifies a parameter that makes fuzzy membership differentiable.

$$f_{bc} = \frac{\left(s(h_b, \mu_c) - \frac{\beta}{\sum_{o=1}^i s(\mu_o, \bar{\mu})} s(\mu_c, \bar{\mu}) \right)^{-1/(t-1)}}{\sum_{o=1}^i \left(s(h_b, \mu_o) - \frac{\beta}{\sum_{o=1}^i s(\mu_o, \bar{\mu})} s(\mu_o, \bar{\mu}) \right)^{-1/(t-1)}}, \quad (27)$$

where β represents the allocated hyper-parameter, and t is fuzzifier.

To facilitate positive fuzzy membership f_{bc} in DNFCs, all η_c is set using $\eta = \frac{\beta}{\sum_{o=1}^i \|\mu_o - \bar{\mu}\|^2}$. This normalization averts numerator to be in negative. Assume within cluster distance $\|h_b - \bar{\mu}_c\|^2$ as x_1 and between-cluster distance $\|\mu_c - \bar{\mu}\|^2$ of low-dimensional feature h_b as x_2 . Here, x_1 seems to acquire exceptionally huge-scale. As training epoch changes, x_1 seems to reduce in small-scale. Here, x_2 can be in huge scale, and thus makes it complex to apply effectual control on x_1 . If $x_1 - x_2 < 0$, the f_{bc} value is not valid, which represent divergent from second constraint objective. Hence, to facilitate x_1 and x_2 wait in the similar scale, x_2 , which is normalized by multiplying $\eta = \frac{\beta}{\sum_{o=1}^i \|\mu_o - \bar{\mu}\|^2}$. Moreover, the correct f_{bc} value is generated by normalizing x_2 , which is capable to evidently imitate sensitivity of distance inconsistency among x_1 and x_2 . Meanwhile, it reveals how between-cluster distance efficiently influences within-cluster distance in DNFCs membership.

The hyper-parameter β is imperative for balancing within-cluster distance and between-cluster distance in cluster space, and it's necessary to facilitate f_{bc} to be positive.

3.2.1 | KL-divergence based objective term

The objective of the second constraint considering clustering layer output is depicted in which DNFCs membership f_{bc} approximate w_{bc} based on KL-divergence.

$$\min KL(Y||Z) = \min \sum_{b=1}^C \sum_{c=1}^i w_{bc} \log \frac{w_{bc}}{f_{bc}}, \quad (28)$$

where $w_{b,c}$ is desired target.

For a gradient descent technique, the gradient and updating weight rule μ_c are obtained. It is observed that optimization of KL divergence makes fuzzy membership f_{bc} to turn out to be as sharp as possible and constricts distribution f_{bc} with all data based on c th cluster,

$$\nabla \mu_{bc} = \frac{x \sum_{b=1}^C \sum_{c=1}^i w_{bc} \log \frac{w_{bc}}{f_{bc}}}{x \mu_{bc}}, \quad (29)$$

$$\mu_{bc} = \mu_{bc} - \alpha \frac{x \sum_{b=1}^C \sum_{c=1}^i w_{bc} \log \frac{w_{bc}}{f_{bc}}}{x \mu_{bc}}, \quad (30)$$

where α refers hyper-parameter, x is distance. When KL-divergence is devised on the basis of loss converged, the weight μ_{bc} connects hidden neuron and fuzzy clustering neuron jointly considering c th cluster. Using DNFCs membership f_{bc} , the objectives of the second constraint attain compactness of within-class distance and separation of between-class distance concurrently.

3.2.2 | Analysis of fuzzifier t

The most imperative part of DNFCs is hyper parameters fuzzifier t . The fuzzifier t manages softness of fuzzy membership, which can control the effect of clustering. Thus, a technique is devised for locating the range of fuzzifier t of fuzzy membership for the deep clustering model. Here, a technique is utilized for approximating range of t , and then select optimum t value based on optimum performance of clustering. Provided $T = \{t_1, \dots, t_b, \dots, t_c\} \in R^x$ as training data, based on the range of fuzzifier attributes t is discovered by $\lambda_{\max}(C_T)$, in which C_T with $\bar{t} = \sum_{b=1}^C t_b / C$ is given as

$$C_T = \sum_{b=1}^C \left((t_b - \bar{t}) (t_b - \bar{t})^T / C \|t_b - \bar{t}\|^2 \right), \quad (31)$$

where $\lambda_{\max}(C_T)$ signifies maximum Eigen value of matrix C_T . Subsequently, the fuzzy parameter range t is expressed as

$$\begin{aligned} t &< (1 - 2\lambda_{\max}(C_T))^{-1}, \text{ If } \lambda_{\max}(C_T) < 0.5, \\ t &\in [1.5, 2.5], \text{ If } \lambda_{\max}(C_T) \geq 0.5. \end{aligned} \quad (32)$$

3.2.3 | Pseudo-labels considering regularization of affinity graph

Assume that features belonging to the same cluster must be close to each other and the significance of discriminative information, a pseudo-labels based affinity regularization is devised.

(a) Graph based regularization on hidden features:

The instinct after graph-based regularization is affinities among input data that must be reliable with ones among new models. Assume $H = \{h_1, \dots, h_b, \dots, h_c\}$, $h \in R^i$ represent encoded features or representations based on input data $T = \{t_1, \dots, t_b, \dots, t_c\}$, $t \in R^x$. Thus, graph-based regularization is given as

$$\min V_h = \min \sum_{b,y=1}^C s(h_b, h_y) s_{by}, \quad (33)$$

where s_{by} , $y = 1, 2, \dots, C$ signifies affinity among data t_b and t_y , and $s(\cdot)$ refers to the affinity function.

Here, a pseudo-labels based technique is utilized that attains discriminative data among fuzzy membership forecasted by clustering layer to evaluate affinity.

On the basis of f_{bc} , or fuzzy membership of data t_b , once can describe pseudo-label v_b of h_b using $v_b = \arg \max f_{bc}$. Here, one can label t_b based on index of transitory maximum membership. Acquiring data t_b and t_y , the affinity s_{by} among representations h_b and h_y is obtained as

$$s_{by} = \begin{cases} \exp(-s(h_b, h_y)/\sigma)/u, v_b = v_y, \\ 0, v_b \neq v_y. \end{cases} \quad (34)$$

Thus, kernel σ is fixed as 1 and u represents the affinity hyper-parameter, which manages the scale of affinity. Moreover, when pseudo-label v_b is altered from labels v_y , the affinity acquires zero value. Thus, a sparse affinity matrix is obtained from provisional membership outputs to avert time-consuming step of discovering KNN.

(b) The pseudo-labels based affinity update:

To improve affinity matrix quality, one needs to update affinity s_{by}^{U+1} using hidden features h_b^U, h_y^U , and pseudo-labels v_b^U, v_y^U from U th training epoch as

$$s_{by}^{U+1} = \begin{cases} \exp(-\|h_b^U - h_y^U\|^2/\sigma)/u, v_b^U = v_y^U, \\ 0, v_b^U \neq v_y^U. \end{cases} \quad (35)$$

Hence, the obtained objective function of the DFC model is expressed as

$$Loss = \sum_{b=1}^C \|e_{f,i}(t_b) - t_b\|_2^2 + \alpha_1 \sum_{b=1}^C \sum_{c=1}^i w_{bc} \log \frac{w_{bc}}{f_{bc}} + \alpha_2 \sum_{b,y=1}^C \|h_b - h_y\|^2 s_{by}. \quad (36)$$

Here, the fuzzy clustering is tuned with the help of the gradient descent algorithm, which is a stochastic and iterative approximation of gradient descent optimization.

4 | RESULTS AND DISCUSSION

The effectiveness of the proposed PRDOA + TDFC is described using dice coefficient, clustering accuracy, Jaccard coefficient, and rand coefficient. The assessment is performed by altering number of slave nodes.

4.1 | Experimental set-up

The execution of the proposed PRDOA + TDFC is performed in Python with PC having Windows 10 OS, Intel i3 core processor and 8GB RAM.

4.2 | Dataset description

The assessment of techniques is performed using two datasets, namely, the COVID-19 machine learning dataset, and analysis of COVID19 disease on Korean dataset.

4.2.1 | Using dataset-1

The COVID-19 machine learning dataset³¹ describes the admission urgency among COVID-19 patients, and it is intended for supervised machine learning or predictive modeling course. This dataset comprises 23 attributes, and the target attribute is urgency_of_admission. The size of overall dataset is 861×23 which contains 861 rows and 23 columns.

4.2.2 | Using dataset-2

The analysis of COVID-19 disease on Korean dataset³² analyzes the COVID-19 disease using Korean dataset. In addition, this dataset utilizes 15 attributes and the target attribute is state. The size of overall dataset is 7382×15 , which contains 7382 rows and 15 columns.

4.3 | Evaluation measures

The performance of the proposed PRDOA + TDFC is evaluated with four measures, such as dice coefficient, clustering accuracy, Jaccard coefficient, and rand coefficient.

(a) Clustering accuracy:

It refers to a metric that evaluates the closeness degree between the estimated value and original value in clustering, and given as

$$\text{Accuracy} = \frac{T^p + T^n}{T^p + T^n + F^p + F^n}, \quad (37)$$

where T^p signifies true positive, F^p denotes false positive, T^n represents true negative, and F^n symbolizes false negative, respectively.

(b) Jaccard coefficient:

It helps evaluate similarity among two different data, and value of the Jaccard coefficient varies among 0 to 100% and formulated as

$$J_c = (S_s) / (S_s + S_d + D_s), \quad (38)$$

where D_s indicates total data points pairs, S_s signifies both data points belong to the same cluster and group, S_d indicates both data points belong to the same cluster, but various groups, and D_s denotes both data points belong to various clusters, but the similar group.

(c) Dice coefficient:

It refers a metric that is utilized to compute similarity among two data samples. The dice coefficient is represented as

$$D_c = \frac{2 |A \cap B|}{|A| + |B|}, \quad (39)$$

where A and B signifies cardinalities of two sets.

(d) Rand coefficient:

It is a measure wherein the percentage of right decisions is made by the algorithm.

$$R_c = \frac{T^p + T^n}{T^p + F^p + F^n + T^n}. \quad (40)$$

4.4 | Performance assessment

The assessment of the proposed PRDOA + TDFC is done using dice coefficient, clustering accuracy, Jaccard coefficient, and rand coefficient using two datasets.

4.4.1 | Assessment with dataset-1

Figure 3 presents assessment of the proposed PRDOA + TDFC with dataset-1 using clustering accuracy, dice coefficient, Jaccard coefficient, and rand coefficient. The assessment of clustering accuracy is displayed in Figure 3A. For number of slaves = 5, the clustering accuracy

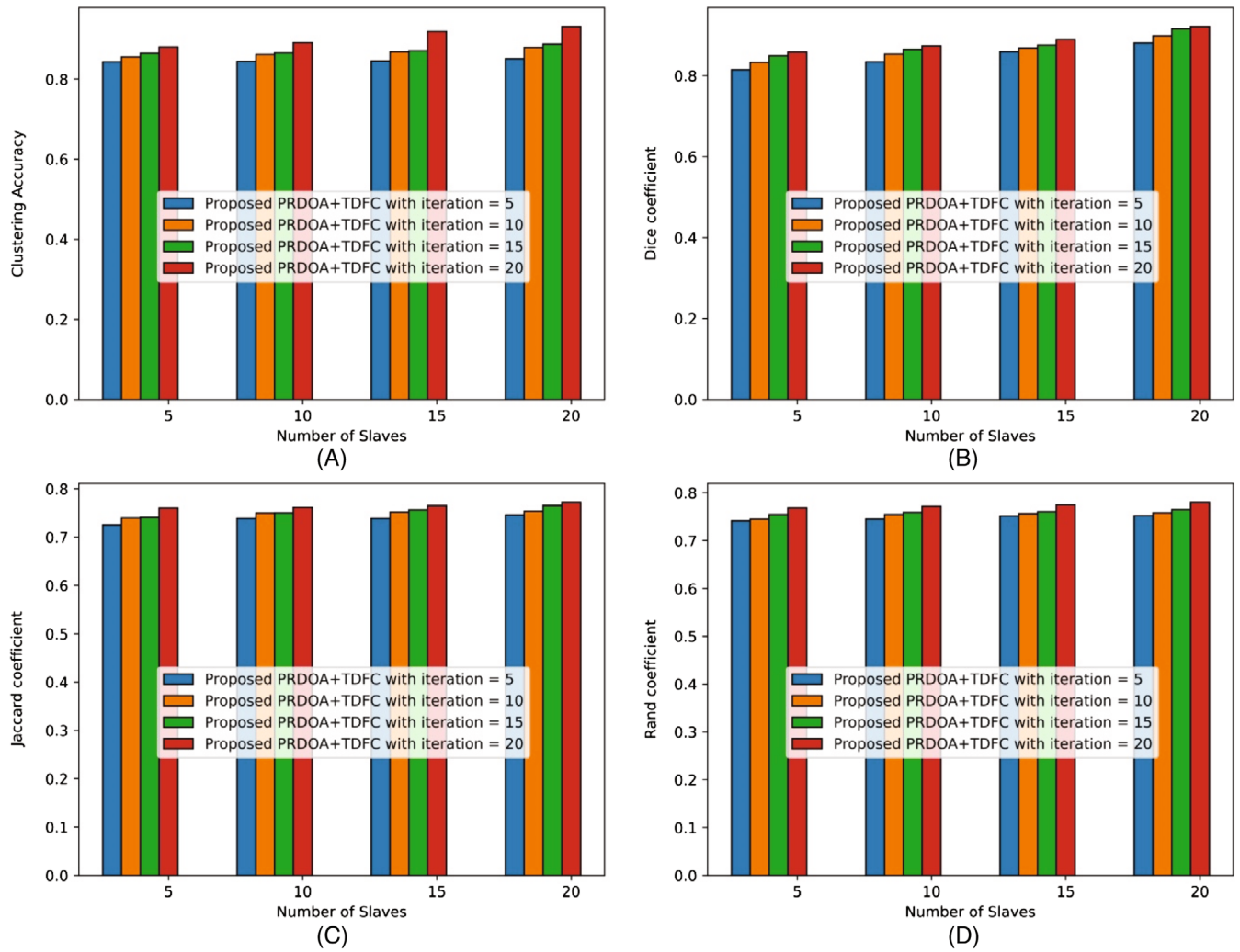


FIGURE 3 Assessment of proposed PRDOA + TDFC with dataset-1 considering. (A) Clustering accuracy, (B) dice coefficient, (C) Jaccard coefficient, and (D) rand coefficient

evaluated by PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.843, 0.856, 0.865, and 0.880. Also, for number of slaves = 20, the clustering accuracy evaluated by proposed PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.851, 0.879, 0.887, and 0.932. The assessment of the dice coefficient is displayed in Figure 3B. For number of slaves = 5, the dice coefficient evaluated by PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.815, 0.833, 0.849, and 0.858. Also, for number of slaves = 20, the dice coefficient evaluated by proposed PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.881, 0.899, 0.916, and 0.922. The assessment of the Jaccard coefficient is displayed in Figure 3C. For number of slaves = 5, the Jaccard coefficient evaluated by PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.726, 0.739, 0.741, and 0.760. Also, for number of slaves = 20, the Jaccard coefficient evaluated by PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.746, 0.754, 0.765, and 0.772. The assessment of the rand coefficient is displayed in Figure 3D. For number of slaves = 5, the rand coefficient evaluated by PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.742, 0.745, 0.755, and 0.768. Also, for number of slaves = 20, the rand coefficient evaluated by PRDOA + TDFC with iteration = 5, 10, 15, and 20 are 0.752, 0.758, 0.765, and 0.781.

4.4.2 | Assessment with dataset-2

The assessment of the proposed PRDOA + TDFC with dataset-2 using clustering accuracy, dice coefficient, Jaccard coefficient, and rand coefficient is depicted in Figure 4. The assessment of clustering accuracy is depicted in Figure 4A. When number of slaves = 5, the clustering accuracy evaluated by proposed PRDOA + TDFC with iteration = 5, 10, 15, and 20 is 0.863, 0.896, 0.897, and 0.907. The assessment of the dice coefficient is displayed

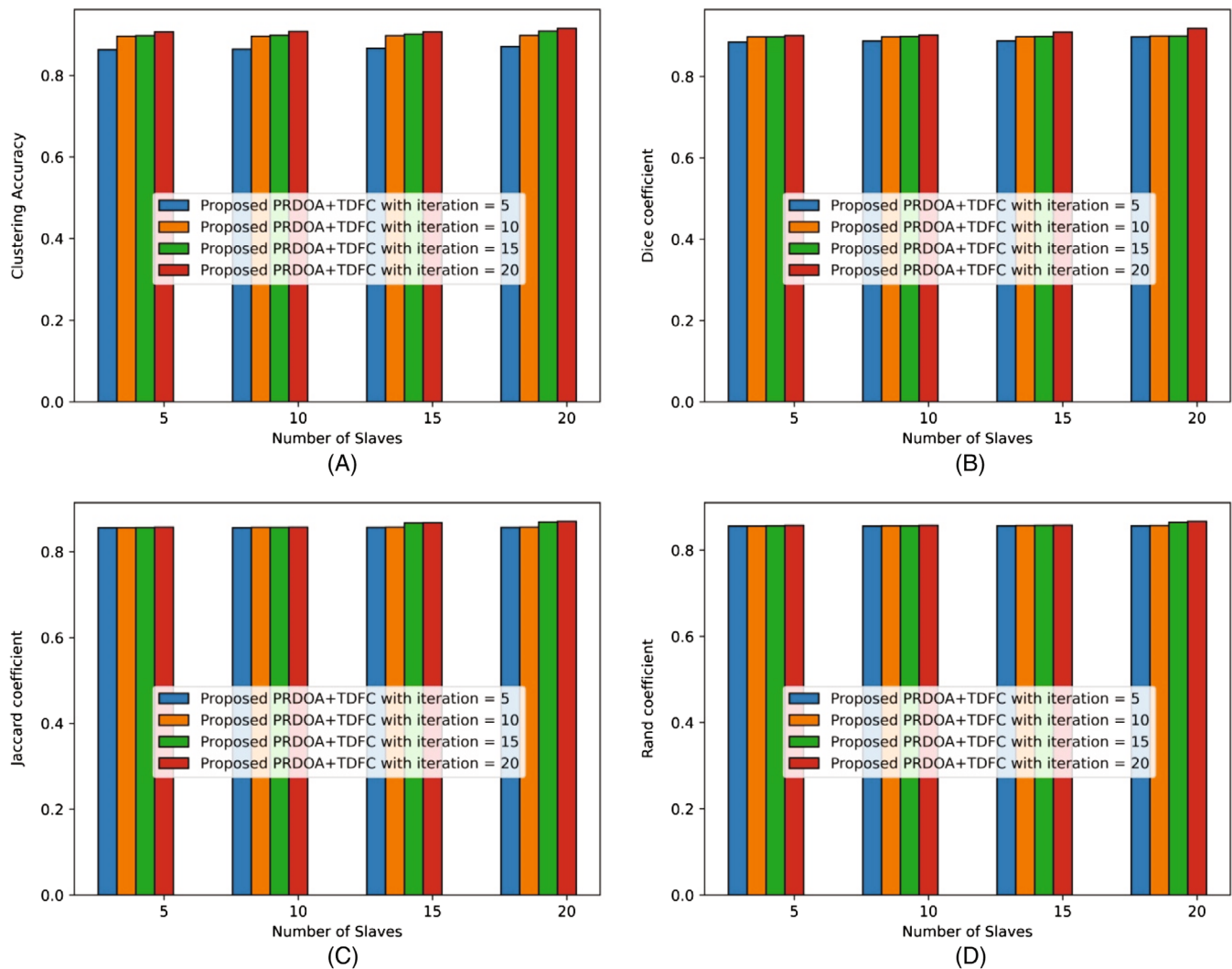


FIGURE 4 Assessment of proposed PRDOA + TDFC with dataset-2 considering. (A) Clustering accuracy, (B) dice coefficient, (C) Jaccard coefficient, and (D) rand coefficient

in Figure 4B. When number of slaves = 20, the dice coefficient evaluated by proposed PRDOA + TDFC with iteration = 5, 10, 15, and 20 is 0.897, 0.900, 0.900, and 0.919. The assessment of the Jaccard coefficient is displayed in Figure 4C. When number of slaves = 5, the Jaccard coefficient evaluated by proposed PRDOA + TDFC with iteration = 5, 10, 15, and 20 is 0.856, 0.856, 0.857, and 0.857. The assessment of the rand coefficient is displayed in Figure 4D. When number of slaves = 20, the rand coefficient evaluated by proposed PRDOA + TDFC with iteration = 5, 10, 15, and 20 is 0.857, 0.857, 0.865, and 0.867.

4.5 | Comparative techniques

The methods taken for assessment are the machine-learning method,²⁶ hierarchical cluster analysis,²⁴ DCNN,²³ K-means clustering method,¹⁴ and proposed PRDOA + TDFC.

4.6 | Comparative assessment

The assessment of techniques is done using dice coefficient, clustering accuracy, Jaccard coefficient, and rand coefficient using two datasets.

4.6.1 | Assessment with dataset-1

Figure 5 presents assessment of techniques with dataset-1 using clustering accuracy, dice coefficient, Jaccard coefficient, and rand coefficient. The assessment of clustering accuracy is displayed in Figure 5A. For number of slaves = 5, the clustering accuracy evaluated by the machine-learning method, Hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.690, 0.706, 0.744, 0.839, and 0.856. Also, for number of slaves = 20, the clustering accuracy evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.712, 0.763, 0.851, 0.856, and 0.879. The performance improvement of the machine-learning method, Hierarchical cluster analysis, DCNN, K-means clustering method with respect to the proposed PRDOA + TDFC using clustering accuracy is 18.99%, 13.19%, 3.185%, and 2.616%.

The assessment of the dice coefficient is displayed in Figure 5B. For number of slaves = 5, the dice coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.709, 0.728, 0.792, 0.815, and 0.833. Also, for number of slaves = 20, the dice coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.737, 0.753, 0.821, 0.875, and 0.899. The performance improvement of the Machine-learning method, Hierarchical cluster analysis, DCNN, K-means clustering method with respect to the proposed PRDOA + TDFC using dice coefficient is 18.02%, 16.24%, 8.676%, 2.669%.

The assessment of the Jaccard coefficient is displayed in Figure 5C. For number of slaves = 5, the Jaccard coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.613, 0.624, 0.693,

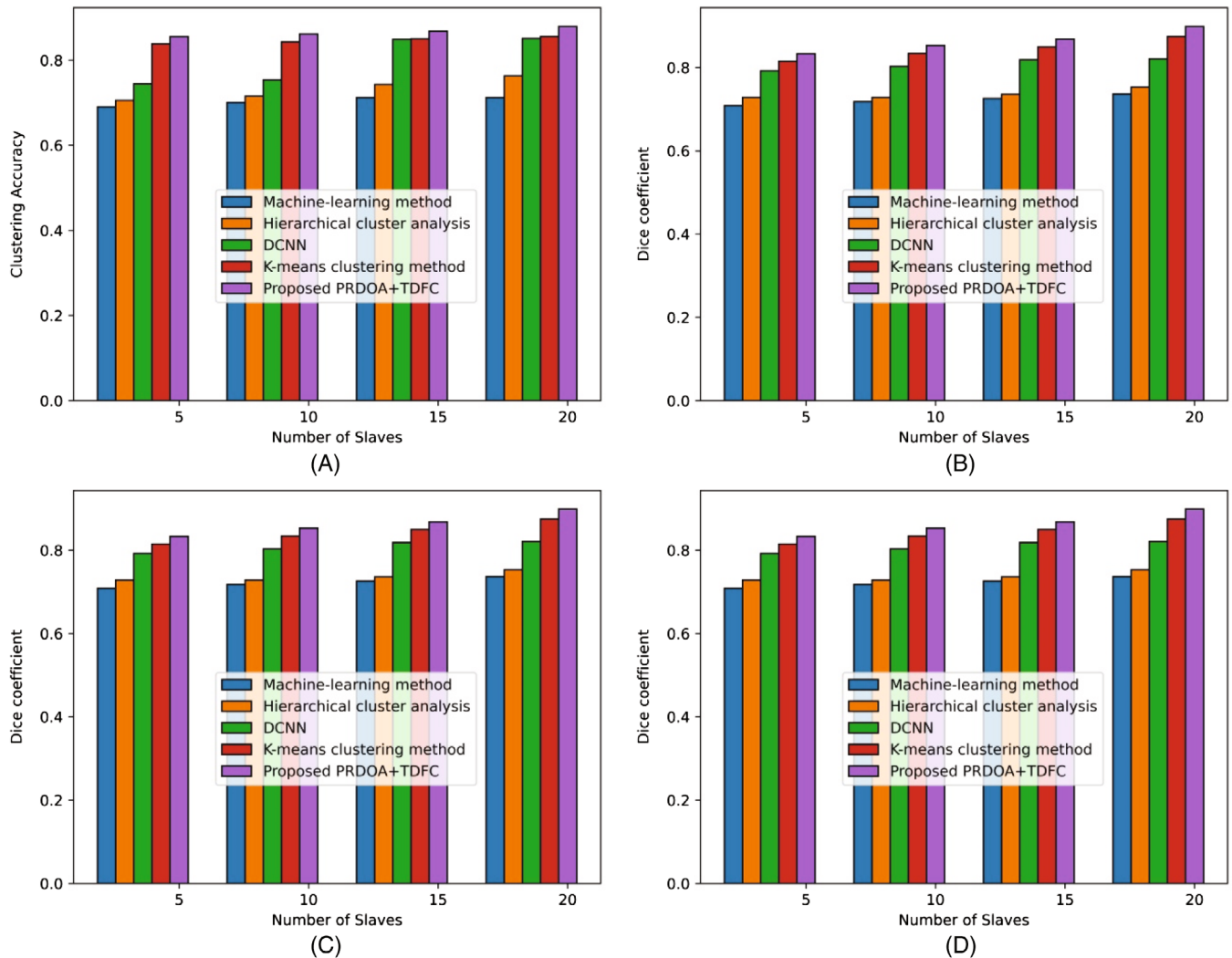


FIGURE 5 Assessment of techniques with dataset-1 considering. (A) Clustering accuracy, (B) dice coefficient, (C) Jaccard coefficient, and (D) rand coefficient

0.722, and 0.739. Also, for number of slaves = 20, the Jaccard coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.634, 0.663, 0.741, 0.748, and 0.754. The performance improvement of the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method with respect to the proposed PRDOA + TDFC using the Jaccard coefficient is 15.91%, 12.06%, 1.724%, 0.795%.

The assessment of the rand coefficient is displayed in Figure 5D. For number of slaves = 5, the rand coefficient evaluated by the Machine-learning method, Hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.599, 0.618, 0.619, 0.729, and 0.745. Also, for number of slaves = 20, the rand coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method, and proposed PRDOA + TDFC are 0.611, 0.635, 0.747, 0.753, and 0.758. The performance improvement of the Machine-learning method, Hierarchical cluster analysis, DCNN, K-means clustering method with respect to the proposed PRDOA + TDFC using rand coefficient is 19.39%, 16.22%, 1.451%, 0.659%.

4.6.2 | Assessment with dataset-2

The assessment of techniques with dataset-2 using clustering accuracy, dice coefficient, Jaccard coefficient, and rand coefficient is displayed in Figure 6. The assessment of clustering accuracy is displayed in Figure 6A. For number of slaves = 5, the clustering accuracy evaluated by the

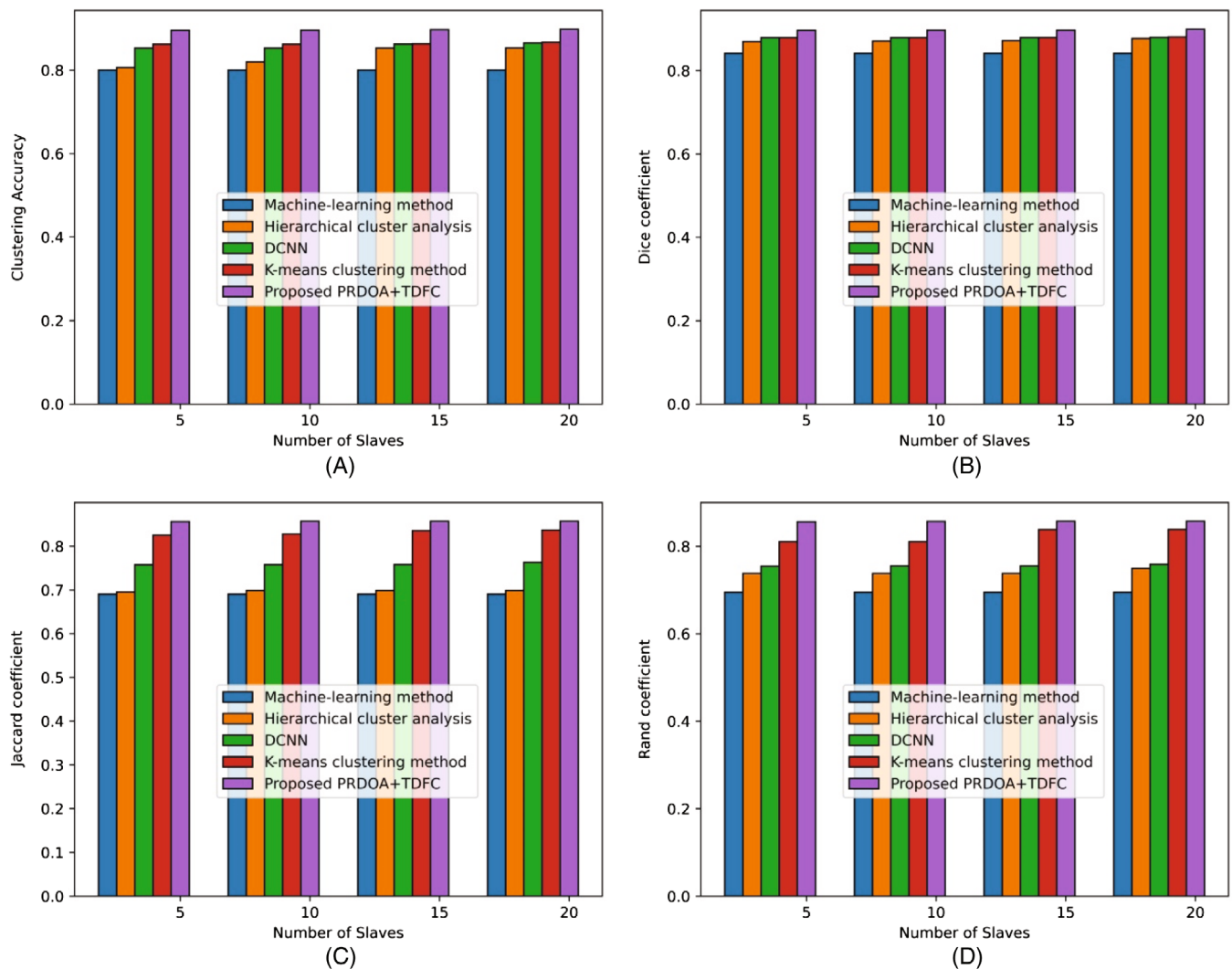


FIGURE 6 Assessment of techniques with dataset-2 considering. (A) Clustering accuracy, (B) dice coefficient, (C) Jaccard coefficient, and (D) rand coefficient

machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.800, 0.806, 0.853, 0.862, while that of the proposed PRDOA + TDFC is 0.896. Also, for number of slaves = 20, the clustering accuracy evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.800, 0.853, 0.865, 0.867, while that of the proposed PRDOA + TDFC is 0.898. The performance improvement of the machine-learning method, hierarchical cluster analysis, DCNN, and K-means clustering method with respect to the proposed PRDOA + TDFC using clustering accuracy is 10.91%, 5.011%, 3.674%, and 3.452%.

The assessment of the dice coefficient is displayed in Figure 6B. For number of slaves = 5, the dice coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.842, 0.870, 0.879, 0.879, while that of the proposed PRDOA + TDFC is 0.897. Also, for number of slaves = 20, the dice coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.842, 0.877, 0.880, 0.881, while that of the proposed PRDOA + TDFC is 0.900. The performance improvement of the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method with respect to the proposed PRDOA + TDFC using dice coefficient is 6.444%, 2.555%, 2.222%, 2.111%.

The assessment of the Jaccard coefficient is displayed in Figure 6C. For number of slaves = 5, the Jaccard coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.690, 0.695, 0.757, 0.825, while that of proposed PRDOA + TDFC is 0.856. Also, for number of slaves = 20, the Jaccard coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.690, 0.699, 0.763, 0.837, while that of the proposed PRDOA + TDFC is 0.857. The performance improvement of the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method with respect to the proposed PRDOA + TDFC using the Jaccard coefficient is 19.48%, 18.43%, 10.96%, 2.333%.

The assessment of the rand coefficient is displayed in Figure 6D. For number of slaves = 5, the rand coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method is 0.695, 0.738, 0.754, 0.810, while that of the proposed PRDOA + TDFC is 0.856. Also, for number of slaves = 20, the rand coefficient evaluated by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.695, 0.749, 0.759, 0.839, while that of the proposed PRDOA + TDFC is 0.857. The performance improvement of the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method with respect to the proposed PRDOA + TDFC using rand coefficient is 18.90%, 12.60%, 11.43%, 2.100%.

4.7 | Comparative discussion

Table 2 describes the comparative assessment of techniques using two datasets. Using dataset-1, the highest clustering accuracy of 0.879 is measured by the proposed PRDOA + TDFC, while the clustering accuracy measured by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method is 0.712, 0.763, 0.851, and 0.856. The highest dice coefficient of 0.899 is measured by the proposed PRDOA + TDFC, while the dice coefficient measured by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method is 0.737, 0.753, 0.821, and 0.875. The highest Jaccard coefficient of 0.754 is measured by the proposed PRDOA + TDFC, while the Jaccard coefficient measured by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.634, 0.663, 0.741, and 0.748. The highest rand coefficient of 0.758 is measured by the proposed PRDOA + TDFC, while the Jaccard coefficient measured by the machine-learning method, hierarchical cluster analysis, DCNN, K-means clustering method are 0.611, 0.635, 0.747, and 0.753. Using dataset-2, the highest clustering accuracy of 0.898, dice coefficient of 0.900, Jaccard coefficient of 0.857, and rand coefficient of 0.857 are measured by the proposed PRDOA + TDFC.

TABLE 2 Comparative assessment

Datasets	Metrics	Machine-learning method	Hierarchical cluster analysis	DCNN	K-means clustering method	Proposed PRDOA + TDFC
Using dataset-1	Clustering accuracy	0.712	0.763	0.851	0.856	0.879
	Dice coefficient	0.737	0.753	0.821	0.875	0.899
	Jaccard coefficient	0.634	0.663	0.741	0.748	0.754
	Rand coefficient	0.611	0.635	0.747	0.753	0.758
Using dataset-2	Clustering accuracy	0.800	0.853	0.865	0.867	0.898
	Dice coefficient	0.842	0.877	0.880	0.881	0.900
	Jaccard coefficient	0.690	0.699	0.763	0.837	0.857
	Rand coefficient	0.695	0.749	0.759	0.839	0.857

5 | CONCLUSION

This article presents to develop a technique, namely, Tanimoto-based deep fuzzy clustering using hybrid optimization for COVID-19 patient analysis using the spark framework. The steps followed for the COVID-19 patient analysis are feature selection and clustering. At first, the input patient's data is splitted into small subsets of data and fed to set several slave nodes. Here, the selection of imperative features is done using the proposed PRDOA in each slave nodes. The proposed PRDOA is obtained by combining DE and PRO. The fitness is newly modeled considering Minkowski distance. The obtained features are merged in master nodes for performing the clustering process. The clustering of patient data is done using the proposed TDFC in the master node. The proposed TDFC is obtained by combining Tanimoto concept and DFC. Thus, the proposed PRDOA and TDFC help to perform COVID-19 patient analysis. The proposed PRDOA with TDFC provided superior performance with the highest clustering accuracy of 89.8%, dice coefficient of 90%, Jaccard coefficient of 85.7% and rand coefficient of 85.7%. In future, other datasets can be utilized to check the feasibility of the technique.

CONFLICT OF INTEREST

Authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in COVID-19 machine learning dataset at https://github.com/ml4lhs/covid19_ml_education³¹ and COVID-19 disease on Korean dataset at <https://github.com/Marcel-Mueck/Covid19PatientsAnalysis>.³²

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